From LLM Cooperation to Coding-Agent Coordination

Weekly Progress: Last Week, Feedback, and This Week

Your Name

Graduate Research Meeting

October 23, 2026

Last Week: Exploring LLM Agent Cooperation

Broad interest: Cooperation among distributed LLM agents

SwarmBench
Local information and collective tasks

Bolzano
Long-running exploration and verification

Remaining issue: The task and research question were still broad.

Background literature: SwarmBench (2025, preprint); Bolzano (2026, preprint).

October 16 Meeting: Feedback

Clear story

Explain context early.

Avoid unnecessary details.

Research evidence

Check related work and benchmarks.

Report time and cost separately.

Time constraint

Define RQ and method within two weeks.

Interim talk: Nov 25–26.

My response: narrow the task and study evaluation methods.

This Week: Focus on Program Repair



Illustrative scenario; no coding-agent prototype or experiments have been completed.

This Week: Three Papers Reviewed

Agentless FSE 2025
Fixed stages: localize, repair, validate

MASAI NeurIPS 2024 Workshop
Specialized sub-agents pass intermediate outputs

MetaGPT ICLR 2024
Roles, standard procedures, and a shared message pool

Agentless [1]; MASAI [2]; MetaGPT [3].

Finding 1: Different Information Flows

System	Coordination	Information flow
Agentless	Fixed workflow	Locate $\rightarrow$ Repair $\rightarrow$ Test
MASAI	Specialized roles	Outputs passed to dependent agents
MetaGPT	Shared message pool	Publish and subscribe to artifacts

They coordinate differently, but solve different tasks.

Summary of architecture descriptions in [1–3]; not a performance ranking.

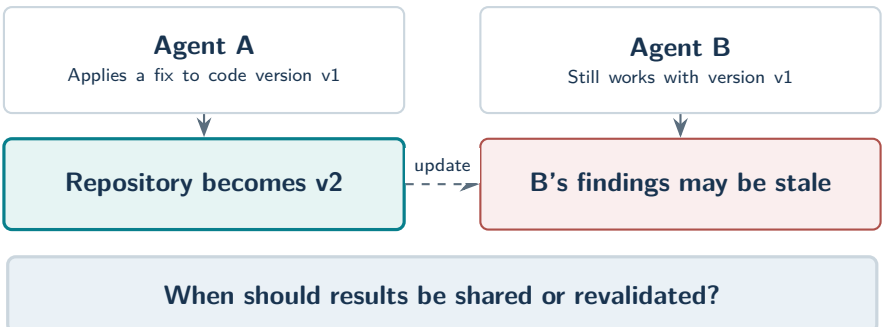
Finding 2: How Existing Work Is Evaluated

Paper	Benchmark	Measurements
Agentless	SWE-bench Lite	Resolved issues, API cost
MASAI	SWE-bench Lite	Resolved issues, localization, cost
MetaGPT	HumanEval / MBPP / SoftwareDev	Pass rate, execution, time, tokens

Lesson: Compare methods on the same task and budget.
Measure success, time, and cost separately.

Benchmarks and metrics reported in [1–3].

A Concrete Coordination Question



Illustrative hypothesis; not an issue experimentally demonstrated by Agentless, MASAI, or MetaGPT.

Candidate Research Questions

RQ1 — Sharing timing

When should parallel coding agents share intermediate results?

RQ2 — Stale information

How does outdated information affect collaborative repair?

Verification before sharing is a related question.

The final RQ has not been selected yet.

Next: Initial Evaluation Design

Task

SWE-bench Lite (small pilot subset)

Measurements

Repair success / Time / Tokens / Messages

Sharing policies

None / Periodic / Version-aware

Controls

Same model and total resource budget

Planned experiment — not implemented yet.

SWE-bench: ICLR 2024 [4]. Comparison policies and metrics are proposals.

Milestones: Two Weeks to Define the Research

When	Deliverable
Oct 16	Meeting feedback received
Oct 17–23	Narrowed task; reviewed three papers
By Oct 30	**Select RQ, related work, method, and benchmark**
Nov 25–26	Interim research presentation



Summary and Questions for Discussion

- **Last week:** Broad investigation of LLM cooperation
- **Feedback:** Need a specific RQ, evidence, and timeline
- **This week:** Program repair and three related papers
- **Next:** Select a coordination question and testbed

Question: Should I focus on sharing timing or stale information?

Is program repair a suitable first evaluation task?

References (Backup)

1. C. S. Xia, Y. Deng, S. Dunn, and L. Zhang.
   Demystifying LLM-based Software Engineering Agents. ACM FSE, 2025.
2. N. Wadhwa et al.
   MASAI: Modular Architecture for Software-Engineering AI Agents. NeurIPS Workshop, 2024.
3. S. Hong et al.
   MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. ICLR, 2024.
4. C. E. Jimenez et al.
   SWE-bench: Can Language Models Resolve Real-World GitHub Issues? ICLR, 2024.
5. *Benchmarking LLMs' Swarm Intelligence (SwarmBench).* arXiv:2505.04364, 2025. (Preprint)
6. *Bolzano: Case Studies in LLM-Assisted Mathematical Research.* arXiv:2604.16989, 2026. (Preprint)